COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 031	**SECTION:**	CPE21S3
DATE PERFORMED:	07/15/26	**DATE SUBMITTED:**	07/15/26
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ACTIVITY NO. 2 [ACTIVITY NAME] INTERACTIVE WIDGET

1. PROCEDURE OUTPUT

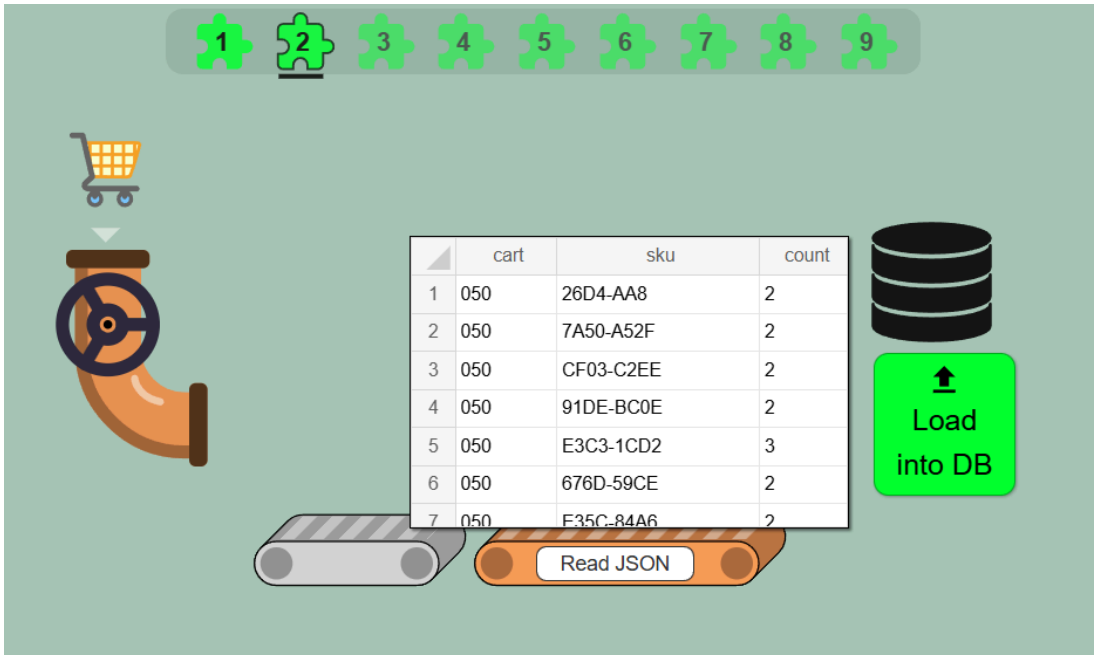
What procedures did you perform?

In this hands-on activity, we first determined the data format by examining its syntax to identify whether it was JSON, XML, or CSV. After identifying the file type, we prepared and cleaned the data before importing it into the database.

For the first dataset, the syntax indicated that it was in XML format. Based on this, I selected the XML option and successfully loaded the data into the database.



For the second dataset, the syntax indicated that it was in JSON format. As a result, I selected the JSON option and successfully imported the data into the database.



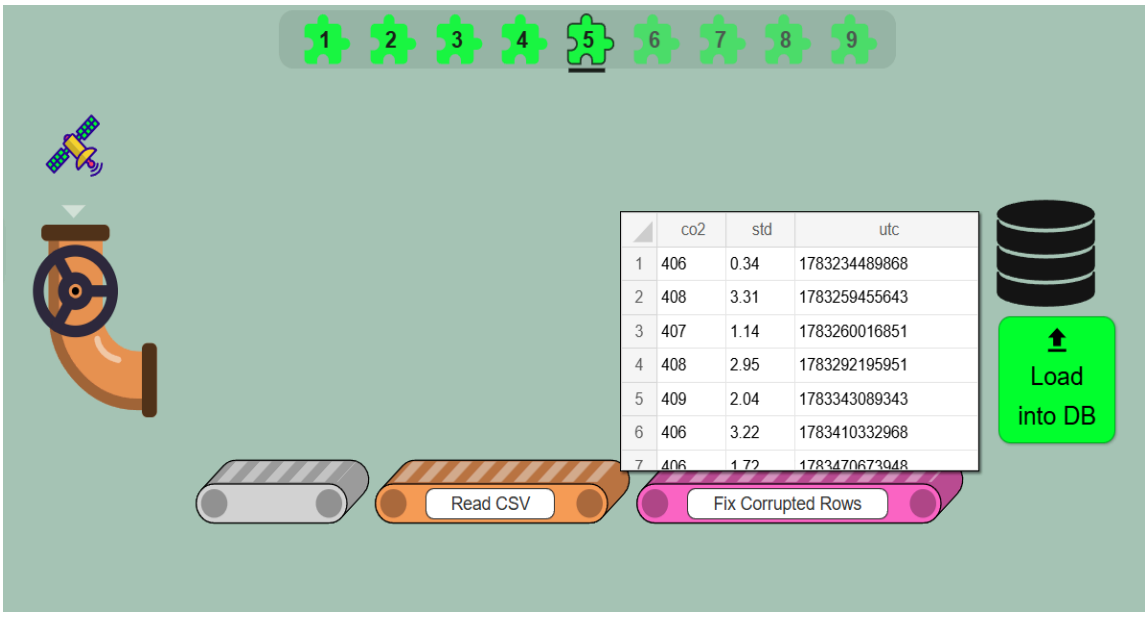
The final dataset was identified as a CSV file because each record is represented on a separate line, with fields separated by commas. I selected the **"Read CSV"** option and successfully imported the data into the database.



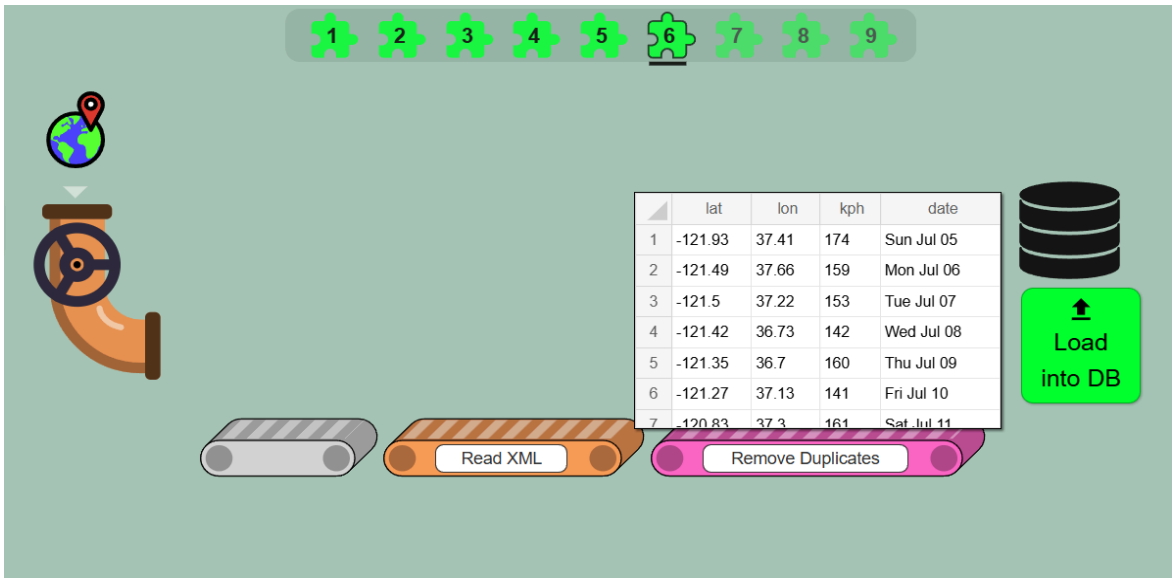
The motion sensor dataset was also identified as a JSON file based on its syntax. However, it contained empty rows, so I removed those rows first to clean the data before importing it into the database.



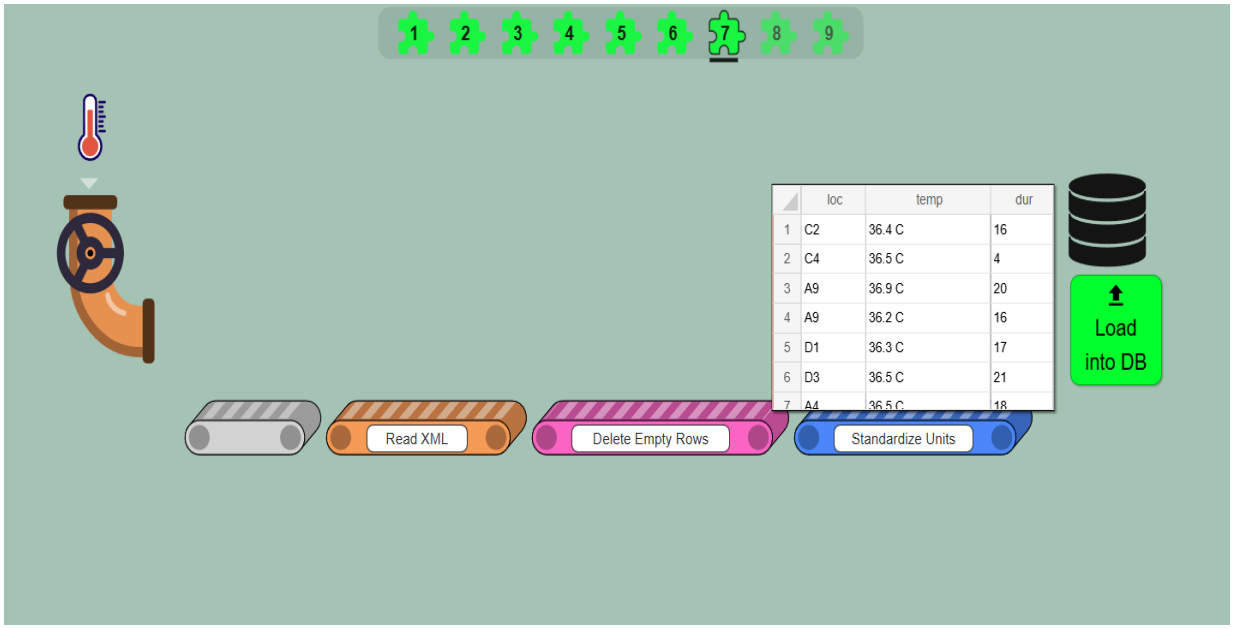
The satellite field dataset was identified as a CSV file, but it contained corrupted rows. I corrected the invalid rows first to ensure the data was accurate and properly formatted. After cleaning the dataset, I successfully imported it into the database.



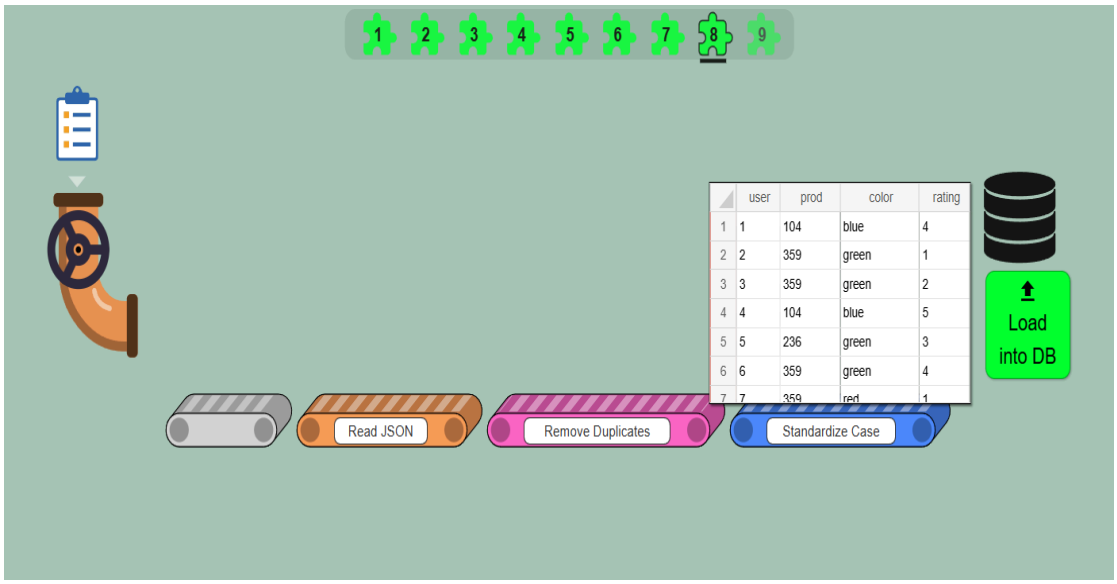
The Global Positioning dataset was identified as an XML file, but it contained duplicate records. I removed the duplicate entries to clean the data before successfully importing it into the database.



The Temperature Ratings dataset was identified as an XML file, but it contained empty rows. I removed the empty rows and standardized the temperature units to ensure consistency. After cleaning and standardizing the data, I successfully imported it into the database.

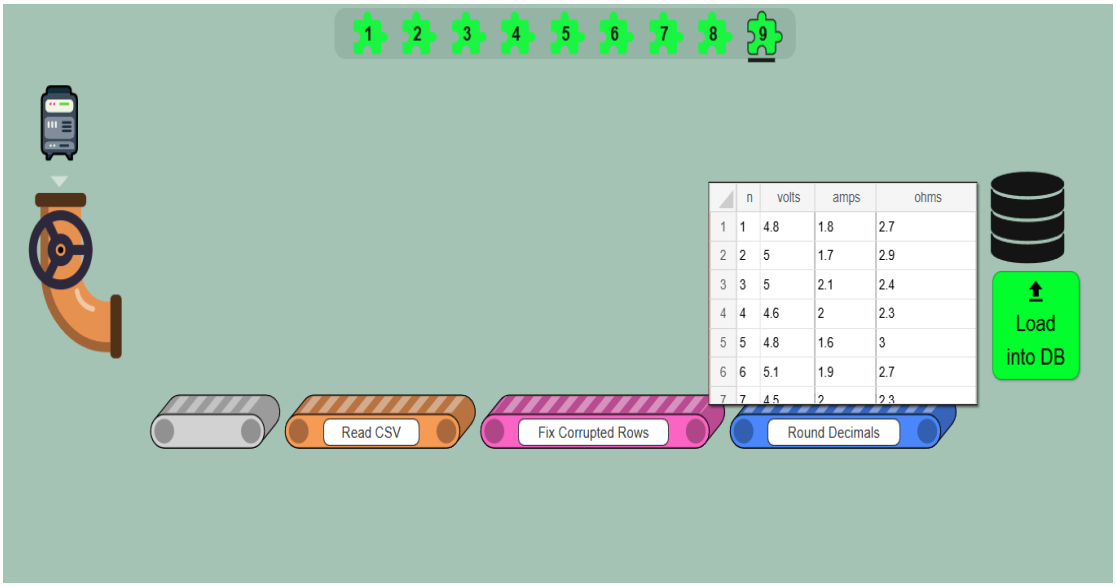


Another dataset contained duplicate records and inconsistent capitalization, with values appearing in both uppercase and lowercase letters. To ensure consistency, I removed the duplicate records and standardized the letter casing before importing the cleaned data into the database.



The Electrical Equipment dataset was identified as a CSV file, but it contained corrupted rows. I corrected the invalid entries to ensure that only accurate data was imported into the database. I also rounded excessively long decimal

values to improve consistency and readability. After cleaning the dataset, I successfully loaded it into the database.



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Conclusion

In this hands-on activity, I successfully identified the format of each dataset, including XML, JSON, and CSV files, by examining their syntax. I cleaned and prepared the data by removing duplicate and empty rows, correcting corrupted records, standardizing values and units, and rounding decimal values when necessary. These data-cleaning steps ensured that the datasets were accurate, consistent, and ready for import. Overall, the activity emphasized the importance of proper data preparation in maintaining data quality and ensuring reliable database management.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction and summarization**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Extent of Use: **first, i use Tagalog language and then translate it into english**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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